

Impact of right heart failure in patients with cardiogenic shock

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Aims

The prognosis of right heart failure (RHF) is purported to be poor, yet there are limited data detailing the implications of concomitant cardiogenic shock (CS) and RHF. We sought to characterize the prevalence and impact of RHF on patients with CS.

Methods and results

We queried the Vizient[®] Clinical Database to identify adults ≥ 18 years old admitted 1 October 2015 to 30 June 2023, with a diagnosis of CS. Using inverse probability of treatment weighting (IPTW), adjusting for demographics, comorbidities, and markers of acuity, we compared in-hospital mortality between patients with and without a diagnosis of RHF as well as by Society of Cardiovascular Angiography and Interventions (SCAI) shock stages B through E. We identified 348 919 patients with CS, including 11.3% ($n = 39\,610$) with RHF. Patients with RHF were younger (60.5 vs. 65.5; $P < 0.001$), more frequently had heart failure (HF) as their primary admission diagnosis (40.8% vs. 23.2%; $P < 0.001$), and presented in earlier SCAI shock stages. Patients with non-RHF CS were more likely to have acute myocardial infarction (AMI) as primary admission diagnosis (20.7% vs. 8.2%; $P < 0.001$) and present with out-of-hospital cardiac arrest (11.5% vs. 5.8%; $P < 0.001$). After adjustment, RHF was associated with a 2.4% [95% confidence interval (CI): 1.9–2.9%; $P < 0.001$] increase in absolute mortality and when stratified by SCAI shock stage. Patients with isolated RHF were found to have a higher absolute in-hospital mortality [9.8%; 95% (CI): 8.6–11.0%; $P < 0.001$].

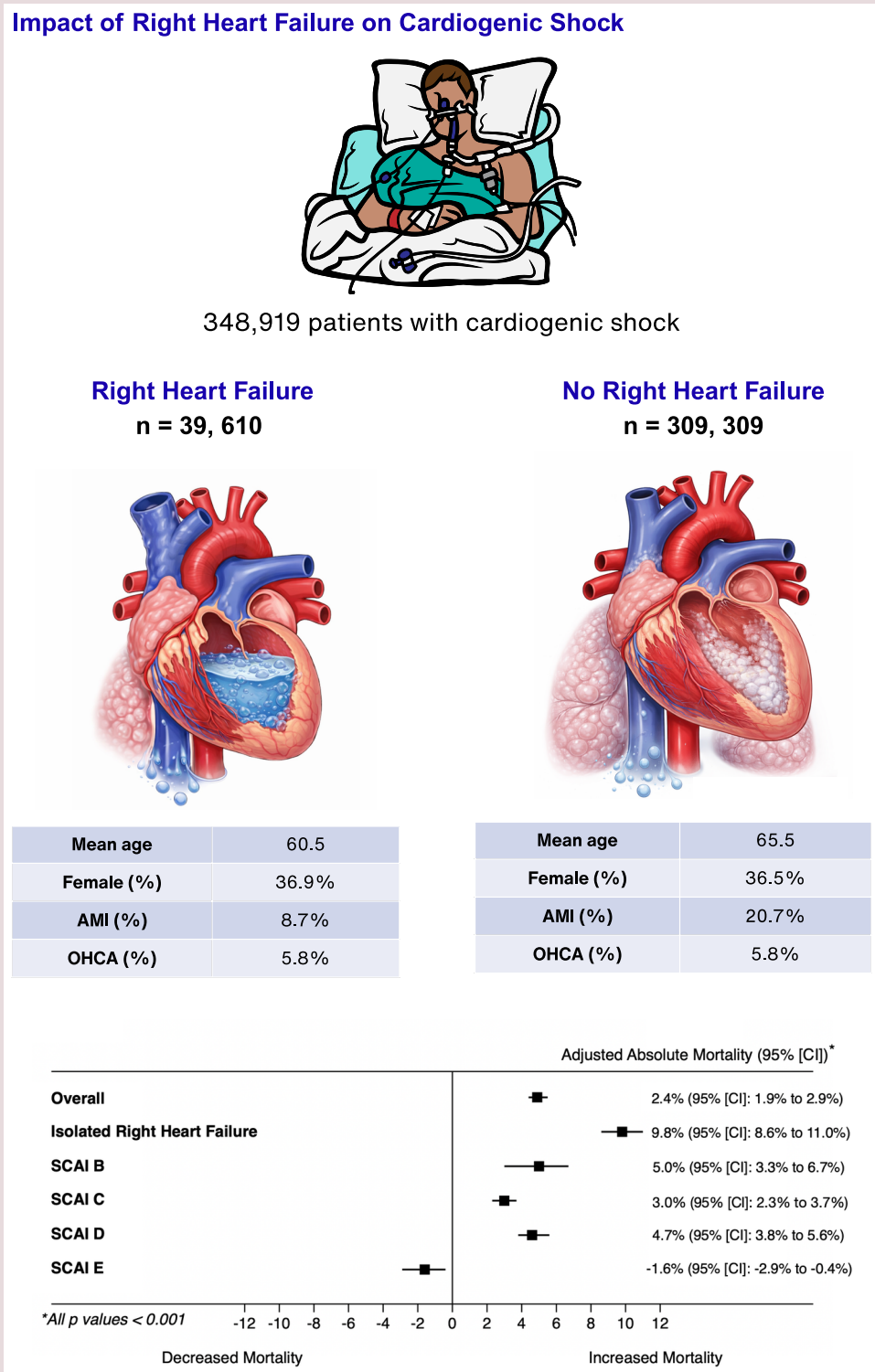
Conclusion

The presence of RHF is associated with increased in-hospital mortality in patients admitted with CS.

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Graphical abstract



SCAI, Society of Cardiovascular Interventions and Angiography

Keywords Right heart failure • Cardiogenic shock • Risk stratification • Outcomes • In-hospital mortality

Introduction

Right heart failure (RHF) remains a complex clinical entity with significant knowledge gaps in its pathophysiology, diagnosis, and management.¹ There is a paucity of guideline recommendations addressing optimal therapeutic strategies in patients with RHF, in part due to the lack of well-performed retrospective and randomized controlled trials in the population.² What guidance does exist typically comes in the form of expert opinion. Contrastingly, appropriate management of left heart failure (LHF) is more clearly defined, including evidence-based recommendations.^{3,4} Patient outcomes in RHF have been historically poor, possibly reflecting this lack of disease-specific recommendations.^{4–6}

Studies have defined the incidence of right ventricular dysfunction in the context of acute myocardial infarction (AMI)-CS to be as high as 38%.⁷ However, the epidemiology of isolated RHF and its impact on CS outcomes remains much less clearly elucidated.⁸ Notably, RHF in CS is an often multifactorial disease process with multiple primary and secondary aetiologies.⁹ This diverse nature further complicates evaluation, diagnosis, and management.

Risk stratification in patients with RHF in CS is an area with limited and contradicting data points. Most existing literature considers the impact of RHF in the context of preexisting left ventricular dysfunction or in specific clinical settings such as AMI.^{7,10,11} Few studies have considered the heterogeneous nature of RHF and its impact on a more diverse CS patient population using real-world data. Therefore, the aim of this study was to better characterize the proportion of patients with concomitant RHF and CS and how RHF impacts outcomes in a large, nationally representative cohort. Additionally, we sought to better characterize the impact of isolated RHF on CS.

Methods

Data source and population

We examined data from the Vizient® Clinical Database, which includes 1350 hospitals from 49 US states; this constitutes over 97% of all academic medical centres as well as their affiliated hospitals. It is comprised of administrative, financial, laboratory, and pharmacy-related inpatient data time-stamped to the day after admission. Inclusion criteria consisted of adults aged ≥18 years, admitted between 1 October 2015 and 30 June 2023, and who had a diagnosis of cardiogenic shock as identified by the International Classification of Diseases, Tenth Revision (ICD-10) code R57.0, which has a high specificity (>95%) and positive predictive value (>90%).¹² To identify the subset of CS patients with RHF, we used ICD-10 codes I50.81 (I50.810, I50.81, I50.812, I50.813, and I50.814) (see [Supplementary material online, Table S1](#)). These were shown to have a positive predictive value of 98% for first-position ICD-10 HF diagnosis and 66% for first/second-position diagnosis in patients with acute heart failure hospitalizations.¹³ We excluded patients transferred from the index hospitalization, with unknown discharge statuses, without medication data, and unknown discharge mortality ([Figure 1](#)). Data were deidentified and were exempt from review by the Yale University Institutional Review Board. Data were used with permission from Vizient, Inc. All rights reserved. All data requests should be made directly to Vizient.

Variables of interest

The demographic variables we queried included age, gender, race, smoking history, and primary payer (commercial, Medicaid, Medicare, other). Comorbidities included a history of dyslipidaemia, heart failure (HF), chronic obstructive pulmonary disease, chronic

liver disease, chronic kidney disease, end-stage renal disease, diabetes, rheumatoid disease, acquired immunodeficiency syndrome (AIDS), previous AMI, previous percutaneous coronary intervention (PCI), previous coronary artery bypass grafting (CABG), history of durable left ventricular assist device (LVAD), and orthotopic heart transplantation. We considered hospital characteristics including region (Midwest, Northeast, South, West), bed-size (<125, 125–300, 301–500, >500), and Association of American Medical Colleges (AAMC)-teaching status.

Identified procedures during hospital admission included non-invasive ventilation, invasive mechanical ventilation (IMV), renal replacement therapy, left heart catheterization, pulmonary artery catheter (PAC) insertion, PCI, intra-aortic balloon pump (IABP), Impella™, extracorporeal membrane oxygenation (ECMO), heart transplantation, and implantation of durable LVAD. ICD-10-CM codes were used to identify procedures (see [Supplementary material online, Table S2](#)).

For each included patient, Society of Cardiovascular Angiography, and Interventions (SCAI) shock stage on admission was calculated utilizing the Critical Care Cardiology Trials Network (CCCTN) approach and the updated 2022 SCAI shock stages.¹⁴

Outcomes

Our primary outcome was in-hospital mortality. Secondary outcomes included total hospital cost as well as total hospital and intensive care unit (ICU) length of stay (LOS) in days.

Statistical analysis

We compared baseline characteristics between patients who had a diagnosis code of RHF and those who did not. Categorical variables were described as frequencies and percentages. We employed *t*-test to compare continuous variables, and the χ^2 test for categorical variables between groups. We used inverse probability of treatment weighting (IPTW), adjusting for demographics (age, sex, race and ethnicity, and primary payer), comorbidities (smoking, coronary artery disease, dyslipidaemia, history of AMI, prior PCI, prior CABG, diabetes, prior heart transplant, prior left ventricular assist device implantation, chronic kidney disease, end-stage renal disease, chronic obstructive lung disease, chronic liver disease, dementia, rheumatic disease, cancer, AIDS, and Charlson comorbidity index), critical admission diagnoses (out-of-hospital cardiac arrest and acute renal failure), any mechanical circulatory support Day 1, any vasoactive medication on Day 1, and hospital/admission characteristics [bed size, region, AAMC-teaching status, and admission timing (weekday vs. weekend)]. Weight stabilization and truncation at the 1st and 99th percentiles were utilized to minimize extreme weights.¹⁵ We targeted a weighted standardized difference <0.10, suggesting good balance between covariates (see [Supplementary material online, Table S3](#)). We used Poisson regression expressed as an incidence rate ratio (IRR) to evaluate hospital and ICU LOS. A gamma regression model was used to evaluate adjusted total hospital costs.

We also conducted several subgroup and sensitivity analyses to further assess the robustness of our findings. First, we excluded patients undergoing CABG, LVAD, heart transplantation, and valve surgery on the first day of admission. We also repeated analyses for any patient that received MCS on the first day of admission, and then stratified by each device (IABP, Impella, and ECMO). Additionally, to reduce the impact of concomitant LHF, we created an 'isolated RHF' subgroup. To create this cohort, ICD-10 codes explicitly identifying RHF (I50.81, I50.810, I50.811, I50.812, and I50.813) were included, and any ICD-10 codes with any mention of LHF were excluded (see [Supplementary material online, Table S1](#)). Of note, for this subgroup, we also excluded ICD-10 code I50.814 (RHF due to LHF), which had previously been included for creation of the RHF cohort during our primary analysis. Next, we repeated our primary analysis stratified by SCAI shock stages B through E. Finally, we conducted *a priori* subgroup analyses amongst only patients with a primary admission diagnosis of AMI, HF, pulmonary embolus (PE), and pulmonary hypertension.

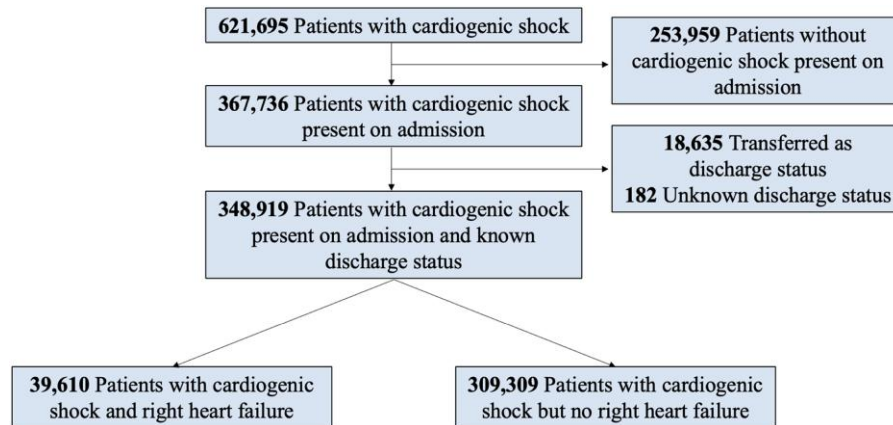


Figure 1 Consort diagram. Graphic representation of study cohort creation.

All analyses were performed on STATA 16.0 (Stata Corp., College Station, TX) with statistical significance considered at a two-tailed $P < 0.05$.

Results

Patient population

Including 348 919 patients with CS, we identified 39 610 (11.3%) with RHF (Figure 1). Patients with RHF were younger than those without RHF (mean age 60.5 vs. 65.5 years; $P < 0.001$), but both groups were similarly male predominant (36.9% vs. 36.5% female, $P = 0.12$) (Table 1). Patients with RHF were more likely to have been admitted to teaching hospitals (78% vs. 63.1%; $P < 0.001$). The non-RHF cohort more frequently had AMI as the admission diagnosis (20.7% vs. 8.2%; $P < 0.001$), including both non-ST elevation myocardial infarction and ST-elevation myocardial infarction, whereas patients with RHF were more likely to have HF (40.8% vs. 23.2%; $P < 0.001$). Notably, patients without RHF presented more frequently with out-of-hospital cardiac arrest (11.5% vs. 5.8%; $P < 0.001$) and had more advanced SCAI shock stages on admission to the ICU (SCAI D-E: 63.6% vs. 54.4%, $P < 0.001$).

Baseline comorbidities are detailed in Table 2. Patients with RHF were more likely to have congestive HF (100% vs. 78.1%; $P < 0.001$), chronic obstructive lung disease (33.1% vs. 27.7%; $P < 0.001$), chronic kidney disease (11.6% vs. 9.3%; $P < 0.001$), and a higher Charlson comorbidity index (4.9 vs. 4.3; $P < 0.001$). Patients without RHF were more likely to have diabetes (43.6% vs. 41.6%; $P < 0.001$), prior AMI (48.9% vs. 34.5%; $P < 0.001$), prior PCI (17.5% vs. 14.7%; $P < 0.001$), and prior CABG (11.0% vs. 9.5%; $P < 0.001$). Patients with RHF were more likely to have undergone LVAD implantation (2.7% vs. 1.4%; $P < 0.001$).

Admission laboratory data

In terms of laboratory values (Table 3), patients without RHF had higher initial lactate values (4.7 mmol/L vs. 4.2 mmol/L; $P < 0.001$) and lower serum pH (7.29 vs. 7.33; $P < 0.001$). Comparatively, we observed that patients with RHF had higher admission creatinine (2.23 mg/dL vs. 2.13 mg/dL; $P < 0.001$), serum blood urea nitrogen (41 U/L vs. 37 U/L; $P < 0.001$), pro-B-type natriuretic peptide (16 554.4 pg/mL vs. 14

814.6 pg/mL; $P < 0.001$), aspartate transaminase (445.8 U/L vs. 334.4 U/L; $P < 0.001$), and total bilirubin (1.9 vs. 1.3; $P < 0.001$).

Critical care procedures and therapies

We also compared the frequency of critical care therapies and procedural interventions (Figure 2) and found that patients with RHF had higher rates of renal replacement therapy (RRT) (5.1% vs. 2.9%; $P < 0.001$), non-invasive ventilation (6.7% vs. 5.1%; $P < 0.001$) and were more likely to require extracorporeal membrane oxygenation (ECMO) (5.1% vs. 2.9%; $P < 0.001$). Conversely, patients without RHF underwent left heart catheterization (LHC) (17.2% vs. 7.6%; $P < 0.001$), PCI (10.3% vs. 3.1%; $P < 0.001$), IABP placement (7.3% vs. 4.6%; $P < 0.001$), CABG (0.8% vs. 0.4%; $P < 0.001$), and IMV (35.0% vs. 25.9%; $P < 0.001$) more frequently. We also observed that a slightly higher proportion of patients without RHF initiated MCS on the first day of admission (11.6% vs. 11.2%; $P = 0.046$), but utilization of Impella was not statistically different between groups (2.9% vs. 3.1%; $P = 0.10$).

Right heart failure mortality in cardiogenic shock

Patients without RHF had a higher unadjusted in-hospital mortality (34.5% vs. 33.0%; $P < 0.001$). However, after IPTW, including excellent covariate balance (see Supplementary material online, Table S3), patients with RHF had a >2% higher absolute risk of in-hospital mortality [2.4% higher mortality rate; 95% confidence interval (CI): 1.9–2.9%; $P < 0.001$] compared to patients without RHF (Figure 3). After exclusion of patients undergoing cardiac surgery on admission, our results were similar [in-hospital mortality 2.2%; 95% (CI): 1.7–2.7%; $P < 0.001$]. Moreover, including only patients that received any MCS on the first day of admission, concomitant RHF was associated with a 9.1% absolute increase in in-hospital mortality [95% (CI): 7.5–10.8%; $P < 0.001$]. Stratified by each MCS device, utilization of IABP [9.7%; 95% (CI): 7.3–12.1%; $P < 0.001$] or Impella [8.7%; 95% (CI): 5.5–12.0%; $P < 0.001$] was associated with a higher absolute in-hospital mortality for patients with concomitant RHF. However, utilization of ECMO was associated with a statistically significant lower mortality [−3.0%; 95% (CI): −5.5%

Table 1 Patient characteristics stratified by diagnosis of right heart failure

	Right heart failure (n = 39 610)	No right heart failure (n = 309 309)	P-value
Demographics			
Age	60.5 (±15.6)	65.5 (±15.0)	<0.001
Female	14 625 (36.9%)	112 958 (36.5%)	0.12
Race			<0.001
Asian	1079 (2.7%)	9219 (3.0%)	
Black	11 326 (28.6%)	63 655 (20.6%)	
Other	3339 (8.4%)	26 487 (8.6%)	
Unknown	1537 (3.9%)	11 449 (3.7%)	
White	22 329 (56.4%)	198 499 (64.2%)	
Primary payer			<0.001
Commercial	8748 (22.1%)	57 850 (18.7%)	
Medicaid	8237 (20.8%)	47 080 (15.2%)	
Medicare	20 200 (51.0%)	184 528 (59.7%)	
Other	2425 (6.1%)	19 851 (6.4%)	
Admission characteristics			
AAMC Teaching Hospital	30 901 (78.0%)	195 272 (63.1%)	<0.001
Region			<0.001
Midwest	10 390 (26.2%)	80 769 (26.1%)	
Northeast	7673 (19.4%)	67 715 (21.9%)	
South	14 367 (36.3%)	110 760 (35.8%)	
West	7180 (18.1%)	50 065 (16.2%)	
Bed size			<0.001
<125	587 (1.5%)	9132 (3.0%)	
125–300	3552 (9.0%)	50 516 (16.3%)	
301–500	7278 (18.4%)	64 223 (20.8%)	
>500	28 193 (71.2%)	185 438 (60.0%)	
Key primary admission diagnoses			
Acute myocardial infarction	3230 (8.2%)	64 077 (20.7%)	<0.001
Heart failure	16 178 (40.8%)	71 606 (23.2%)	
Pulmonary hypertension	1131 (2.9%)	541 (0.2%)	
Pulmonary embolism	753 (1.9%)	3012 (1.0%)	
Admission SCAI shock stage			
B	1458 (3.9%)	5711 (1.9%)	<0.001
C	15 678 (41.7%)	102 324 (34.5%)	
D	12 932 (34.4%)	116 901 (39.4%)	
E	7521 (20.0%)	71 664 (24.2%)	

Data are presented as mean (± standard deviation) for continuous measures and n (%) for categorical variables.
AAMC, Association of American Medical Colleges; SCAI, Society of Coronary Angiography and Intervention.

to −0.5%; $P = 0.018$). In sensitivity analysis, patients with isolated RHF had a 9.8% [95% (CI): 8.6–11.0%; $P < 0.001$] higher absolute in-hospital mortality compared to patients without RHF. When stratified by SCAI shock stage, patients with RHF were associated with a higher in-hospital mortality for stages B through D but not E (Figure 3). Finally, when stratified by primary admission diagnoses, a primary diagnosis of AMI or HF was associated with a higher in-hospital mortality but not PE or pulmonary hypertension (Figure 4).

Secondary outcomes

Patients with RHF had longer ICU LOS (6 days vs. 3 days; $P < 0.001$), total hospital LOS (12 days vs. 7 days; $P < 0.001$), and higher average admission costs (\$474 695 vs. \$30 047; $P < 0.001$). After multivariable adjustment, patients with RHF similarly had a longer ICU LOS [IRR: 1.42; 95% (CI): 1.41–1.42; $P < 0.001$], total hospital LOS [IRR: 1.29; 95% (CI): 1.29–1.30; $P < 0.001$], and higher mean difference in total hospital costs [\$23 532; 95% (CI): \$22 604–24 460; $P < 0.001$].

Table 2 Comorbidities stratified by diagnosis of right heart failure

	Right heart failure (n = 39 610)	No right heart failure (n = 309 309)	P-value
Comorbidities			
Charlson comorbidity index	4.9 (±2.6)	4.3 (±2.6)	<0.001
Prior AMI	13 683 (34.5%)	151 300 (48.9%)	<0.001
Coronary artery disease	18 232 (46.0%)	173 285 (56.0%)	<0.001
Prior PCI	5811 (14.7%)	54 160 (17.5%)	<0.001
Prior CABG	3771 (9.5%)	34 069 (11.0%)	<0.001
Dyslipidaemia	17 266 (43.6%)	150 139 (48.5%)	<0.001
Smoking	20 312 (51.3%)	152 582 (49.3%)	<0.001
Congestive heart failure	39 610 (100.0%)	241 682 (78.1%)	<0.001
COPD	13 108 (33.1%)	85 597 (27.7%)	<0.001
Chronic kidney disease	4600 (11.6%)	28 728 (9.3%)	<0.001
ESRD	3642 (9.2%)	23 645 (7.6%)	<0.001
Rheumatoid disease	1877 (4.7%)	9569 (3.1%)	<0.001
Diabetes	16 474 (41.6%)	134 711 (43.6%)	<0.001
Chronic liver disease	7105 (17.9%)	30 823 (10.0%)	<0.001
AIDS	323 (0.8%)	1506 (0.5%)	<0.001
Prior LVAD	1066 (2.7%)	4478 (1.4%)	<0.001
Prior heart transplant	46 (0.1%)	324 (0.1%)	0.51

Data are presented as n (%) for categorical variables.

AMI, acute myocardial infarction; PCI, percutaneous coronary intervention; CABG, coronary artery bypass grafting; ESRD, end-stage renal disease; COPD, chronic pulmonary obstructive disease; AIDS, acquired immunodeficiency syndrome; LVAD, left ventricular assist device.

Table 3 Admission laboratory values in patients stratified by diagnosis of right heart failure

	Right heart failure (n = 39 610)	No right heart failure (n = 309 309)	P-value
Lactate (mmol/L)	4.2 (± 3.9)	4.7 (± 4.3)	<0.001
Creatinine (mg/dL)	2.23 (± 1.70)	2.13 (± 1.87)	<0.001
eGFR (mL/min/1.73 m ²)	43 (± 24)	45 (± 25)	<0.001
Blood urea nitrogen (U/L)	41 (± 27)	37 (± 27)	<0.001
proBNP (pg/mL)	16 554 (± 21 782)	14 815 (± 23 362)	<0.001
pH	7.33 (± 0.14)	7.29 (± 0.16)	<0.001
Bicarbonate	22 (± 6.2)	21 (± 6.0)	<0.001
Aspartate transaminase (U/L)	446 (± 1473.9)	334 (± 1150.8)	<0.001
Alanine aminotransferase (U/L)	288 (± 802)	217 (± 665)	<0.001
Total bilirubin (mg/dL)	1.9 (± 2.3)	1.3 (± 1.8)	<0.001
Sodium	135 (± 6)	137 (± 6)	<0.001
White blood cell count	11.4 (± 8.1)	12.6 (± 10.0)	<0.001
Haemoglobin	12.0 (± 2.7)	11.9 (± 2.7)	<0.001
Platelets	203 (± 101)	220 (± 104)	<0.001

Data are presented as mean (± standard deviation) for continuous measures.

eGFR, estimated glomerular filtration rate; proBNP, pro-B-type natriuretic peptide.

Discussion

In this analysis of a multicentre, nationally representative database (including most teaching hospitals in the USA), we found

that concomitant RHF was present in 11.3% of patients with CS and had important prognostic implications. Patients with RHF were ~5 years younger, and presented with lower SCAI shock stage and lower serum lactate than did patients without

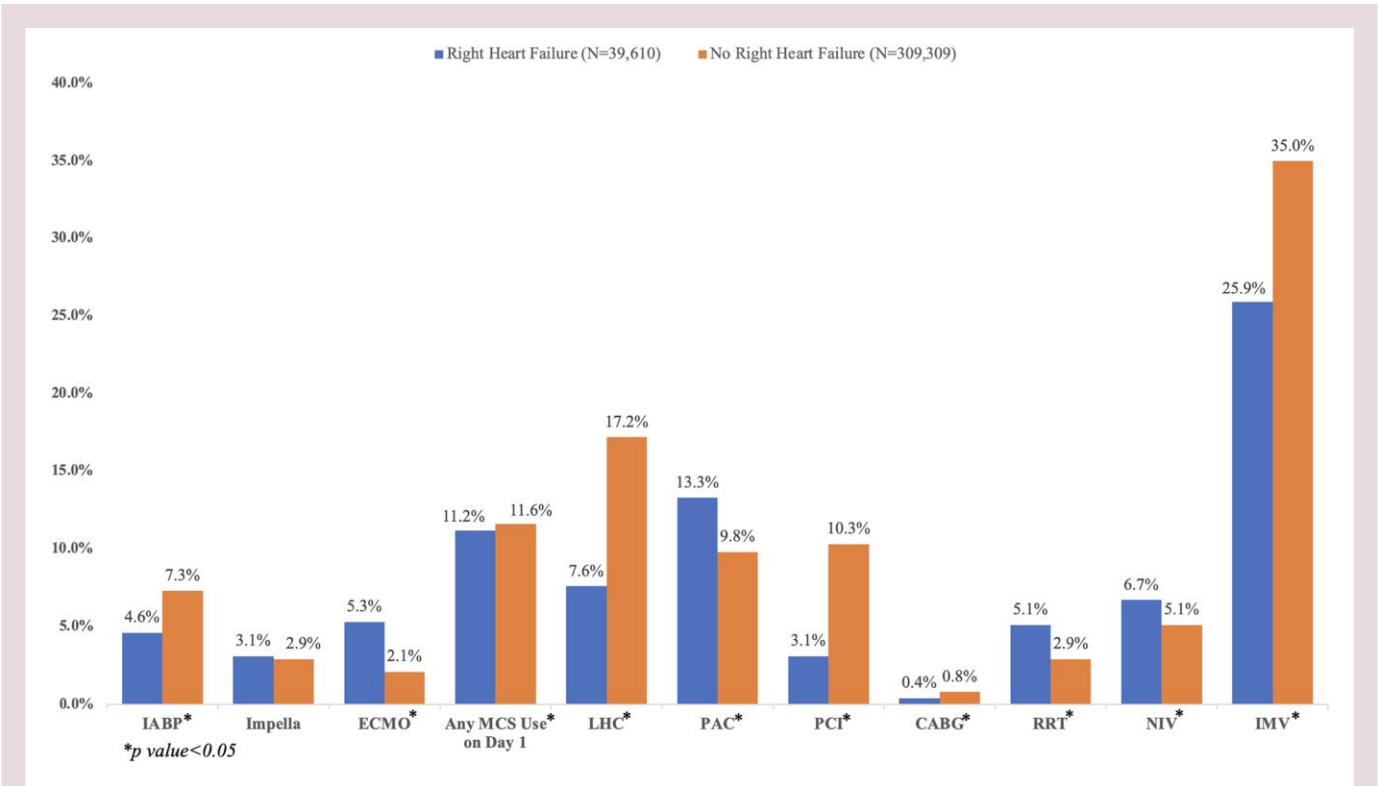


Figure 2 Critical care therapies and interventions on day of admission stratified by diagnosis of right heart failure. ICD-10-CM codes were used to identify procedures on the day of ICU admission and were compared between the RHF cohort and the non-RHF cohort. IABP, intraortic balloon pump; ECMO, extracorporeal membrane oxygenation; MCS, mechanical circulatory support; LHC, left heart catheterization; PAC, pulmonary artery catheterization; PCI, percutaneous coronary intervention; CABG, coronary artery bypass grafting; activator; RRT, renal replacement therapy; NIV, non-invasive ventilation; IMV, invasive mechanical ventilation.

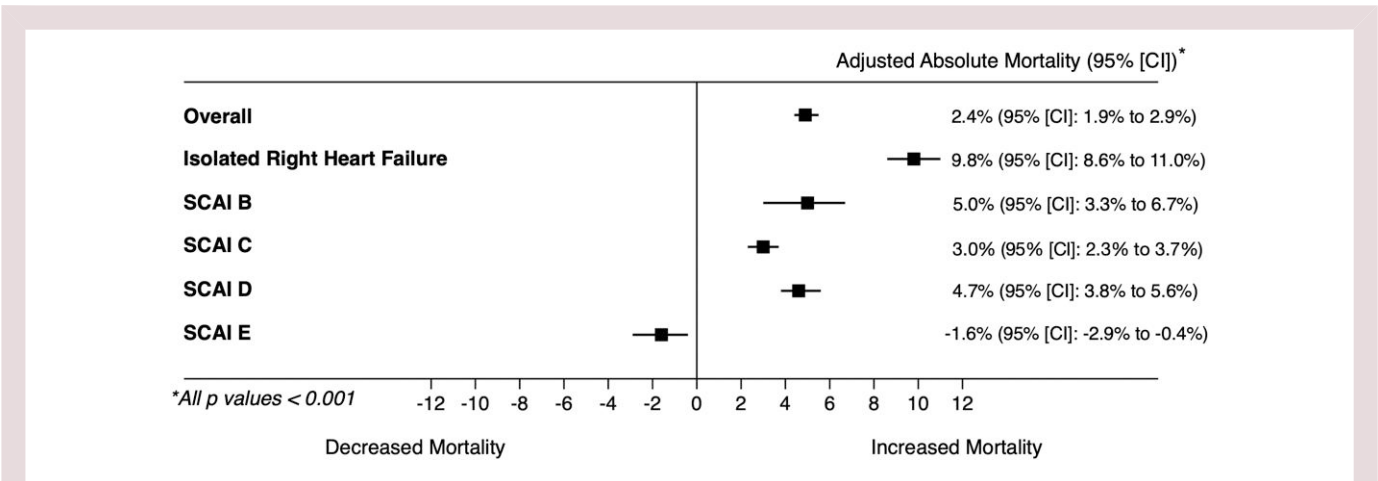


Figure 3 In-hospital mortality in patients with cardiogenic shock stratified by diagnosis of right heart failure. We compared in-hospital mortality patients with CS with and without RHF and stratified them based on SCAI shock stages. SCAI, Society of Cardiovascular Interventions and Angiography.

RHF. These factors may have influenced unadjusted in-hospital mortality analyses, leading to lower mortality in the RHF group. However, after multivariable adjustment, the presence of concomitant RHF in patients with CS was associated with a significant, 2.5% absolute increase in in-hospital mortality. This

increased risk was consistent across presenting SCAI shock stages B through D and was most pronounced in patients with isolated RHF (approximately 10% absolute risk increase). Finally, RHF was associated with longer ICU and hospital LOS and higher total hospital costs. Taken together, RHF occurs in

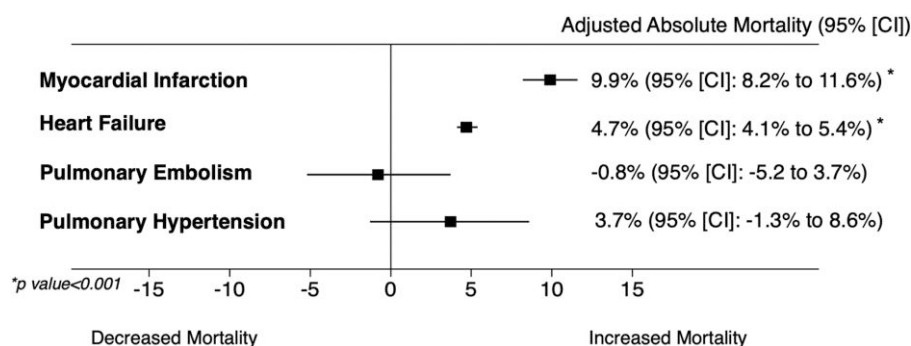


Figure 4 In-hospital mortality in patients with cardiogenic shock stratified by principal admission diagnosis.

over 1 in 10 patients with CS and is associated with significant morbidity and mortality.

Our results are supported by several studies finding worse outcomes in patients with RHF.^{7,16,17} The higher mortality after adjustment in our study is likely multifactorial. The right ventricle has a complicated physiology often impacted by chronicity of the dysfunction, specific cause for RHF (e.g. increases in right ventricular afterload), and concomitant left ventricular involvement.^{2-4,18} Furthermore, validated treatment options for right-sided dysfunction are practically non-existent.¹⁹⁻²¹ Our finding that isolated RHF was associated with the greatest increase in mortality may further support the specific challenges of predominant right ventricular dysfunction. In addition to finding significant differences in MCS use on admission between those with and without RHF, other critical care therapies, such as pulmonary artery catheterization, non-invasive, and IMV further highlight the unique patient population identified and the lack of effective early medical and procedural interventions.

The prevalence of RHF defined in our research aligns with previous studies that used invasive haemodynamics to define isolated RHF.¹⁷ However, the reported prevalence of RHF in CS varies substantially in the existing literature. One study reported rates of RHF around 37% in AMI-CS, with 5.3% of patients having a RHF predominant CS phenotype and 5.5% having CS following isolated right ventricular infarction.²² Another registry found the prevalence of RHF to be close to 44% in AMI-CS and, although not specifically defined, RHF was estimated to be even more prevalent in non-AMI CS patients.^{17,23} Note that the criteria employed to identify RHF differed significantly between studies and can account for the variabilities observed in the existing literature.^{17,22-24} These discrepancies may also be due to increased inter-operator variability in RHF-specific ICD code reporting. Additionally, given that right ventricular dysfunction commonly leads to impairment of left ventricular function, patients identified as having RHF could be characterized as having biventricular HF, which is independently associated with poor outcomes.^{25,26}

Our RHF cohort also had distinct comorbidities, laboratory profiles, and underwent unique critical care therapies and interventions compared to those without RHF. Those with CS and concomitant RHF were characterized by having more comorbidities despite a lower age, less likely to have coronary disease or previous coronary interventions, but more likely to have non-cardiac chronic medical diagnoses, such as chronic lung, kidney, and liver disease. Regarding labs, patients with RHF had higher liver enzymes, total bilirubin, and creatinine on admission, suggestive of higher end-organ involvement and potentially increased venous

congestion compared to those without concomitant RHF. Finally, those with concomitant RHF more commonly required RRT, NIV, and biventricular support with ECMO on the first day of admission. Taken together, patients with CS and RHF represent a unique shock phenotype that requires further investigation.

Patients in our RHF cohort had significantly higher hospitalization costs when compared to those without RHF, likely partly driven by the longer ICU and total hospital LOS, which persisted despite a higher adjusted mortality. Another explanation could be higher use of ECMO, which has been reported to be exponentially more costly than other MCS options.²⁷ Given the costliness of ECMO and that other less expensive/costly MCS options are often inappropriate for RHF, this also highlights the importance of research into and the development of medical management strategies in RHF, to avoid the need for MCS escalation. We are unaware of other studies detailing the financial impact of RHF complicated CS; thus, highlighting further novelty of our study.

Our study should be interpreted with consideration of several important limitations. First, this was a retrospective analysis using an administrative database. Validation of ICD-10 codes to identify patients with RHF is limited, and we lack important information that the clinical team presumably used to diagnose RHF and to define the severity of right ventricular dysfunction, such as echocardiographic and invasive haemodynamic data. Moreover, relying on ICD coding for the identification of patients with RHF inherently introduces diagnostic imprecision to the results of our study. In addition, RHF was rarely ($n = 491$, 0.14%) the primary admission diagnosis. However, our main goal was not to study primary RV failure but the impact of concomitant RHF. Second, we recognize RHF is a complex clinical entity arising from multiple aetiologies, and although we performed subgroup analyses of common acute and chronic types of RHF, we are unable to identify the chronicity of right heart dysfunction, which is important for treatment and prognostication. Third, despite excellent covariate balancing and several sensitivity analyses, there are certainly additional unmeasured confounders that were not accounted for in our analysis.

In conclusion, we found that RHF occurred in over 10% of patients with CS. After adjustment, patients with concomitant RHF had an increased in-hospital mortality, which was most pronounced in patients with isolated RHF. Given the worst outcomes, unique pathophysiology of RHF, and limited treatment options, our work highlights the need for further research on dedicated therapies and treatment approaches for this unique patient population. Future studies are required to validate our findings of RHF as a risk marker for worse outcomes.

Supplementary material

Supplementary material is available at [European Heart Journal: Acute Cardiovascular Care](#) online.

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